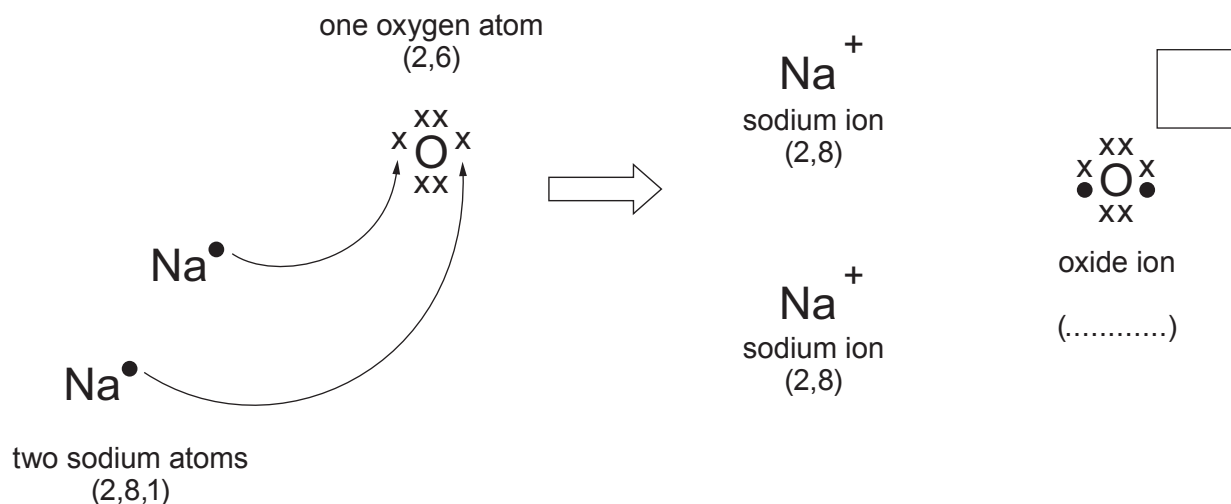


4. (a) The diagram shows the electronic changes that occur when sodium reacts with oxygen to form sodium oxide. The ● and x symbols are outer shell electrons.



- (i) **Complete the diagram** by putting in the electronic structure **and** the charge of the oxide ion. [2]
- (ii) Complete and balance the **symbol** equation for the reaction between sodium and oxygen. [2]



- (b) The table shows the electronic structure of the elements present in water and hydrogen chloride.

Element	Electronic structure
hydrogen	1
oxygen	2,6
chlorine	2,8,7

The diagram shows the bonding in a water molecule.



Give the **letter** of the diagram which shows the bonding in a hydrogen chloride molecule. [1]



A



B



C

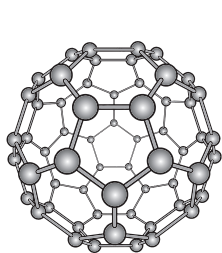


D

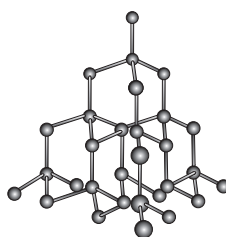
Letter



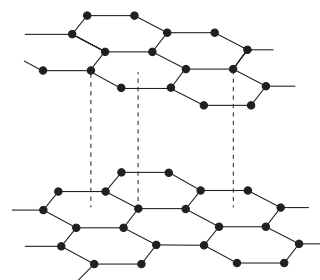
(c) Fullerene, diamond and graphite are different forms of carbon.



fullerene



diamond



graphite

(i) Underline the structure of these three forms of carbon.

[1]

simple molecular

giant covalent

giant ionic

(ii) The box shows some properties of different forms of carbon.

transparent	insoluble in water	conducts electricity
hollow	soft	does not conduct electricity
		hard

Complete the table by choosing the property which makes the different forms of carbon suitable for the uses shown. [3]

Form of carbon	Use	Property
diamond	drill bits	
graphite	pencils	
fullerene	drug delivery	

